

$$\frac{13}{6}$$

Set 4

MISS Sample Test

confusion matrix

check
note

$$F1 \text{ score} = \frac{2PR}{P+R}$$

$$\frac{\mu - \bar{x}}{\frac{s}{\sqrt{n}}}$$

$$n = 11$$

$$p = 0.5$$

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6. Box has 3 Blue balls
 4 green
 5 yellow } 12 balls
 3 without replacement

Total outcomes $12C_3 = 220$

Favourable Atleast 2 same color
 $\Rightarrow$

2 same color
 or 3 same color
 shortcut:

$$P(\text{atleast 2 same}) = 1 - P(\text{all different})$$

$$\cancel{220} (3C_1 \times 4C_1 \times 5C_1) = 60$$

ways ways ways

60 ways different color
 can be chosen

$$220 - 60 = 160 \text{ favorable outcomes}$$

$$\frac{160}{220} = \frac{8}{11}$$

15. ~~2 pair dice rolled.~~

15. 2 fair dice

$$P(\text{sum} \neq 4) = 0$$

Total outcome: $6^2 = 36$

case 1: sum is 4

$\left. \begin{array}{l} (1, 3) \\ (2, 2) \\ (3, 1) \end{array} \right\} 3$

case 2: sum is 8:

$\left. \begin{array}{l} (2, 6) \\ (3, 5) \\ (4, 4) \\ (5, 3) \\ (6, 2) \end{array} \right\} 5$

case 3: sum is 12

$(6, 6) \rightarrow 1$

Total:

Favorable outcomes 9

$$P = \frac{F}{T} = \frac{9}{36} = \frac{1}{4}$$

17. $f(x) = \underbrace{x^2}_u \underbrace{\sin x}_v$

$$f'(x) = x^2 \cos x + 2x \sin x$$

$$f''(x) =$$

$$2x \cos x + x^2(-\sin x) +$$

$$x (x \cos x + 2 \sin x)$$

$$f''(x) =$$

$$2x \cos x + x^2 (-\sin x)$$

$$+ 2 \sin x + 2x \cos x$$

$$\equiv 4x \cos x - x^2 \sin x + 2 \sin x$$

Set 5

2024 slot 3

1) Mode when frequency interval is given

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times$$

highest
band's
lowest
limit

preceeding
successor